Chapter 5

Rare Cell Type Detection

Lan Jiang

Abstract

High-throughput single-cell technologies have great potential to discover new cell types. Here, we present a novel computational method, called GiniClust (Jiang et al., Genome Biol 17(1):144, 2016), to overcome the challenge of detecting rare cell types that are distinct from a large population.

Key words Clustering, Single-cell analysis, RNA-seq, qPCR, Gini index, Rare cell type

1 Introduction

The cell is the basic unit of structure and function in life; however, our knowledge of cell types remains largely incomplete. Interestingly, in many development and disease context, rare cell types often been see to play an important role although they only contribute to a small proportion of the cell population. For example, stem cells that contribute to new born neuron in adult brain are critical to reverse neurodegenerative diseases [1], and drug-resistant cells are the key barrier to cure cancer [2].

Genome-wide gene expression profiles now are widely accepted to define cell types. Recently technical advance of massive parallel single-cell RNA-seq on large-scale provides an unprecedented opportunity to discover previously unrecognized cell types due to their rarity. Although the number of cells being profiled single-cell transcriptome assay increase the chance that rare cell being sampled, tailored computational method to detect them remains highly demanded.

One of the major challenges is to identify genes that are associated with rare cell types without prior biological knowledge. GiniClust [3] adapted Gini index, an informative statistical measure widely used in social domain, to selecting rare cell-type-associated genes. It is implemented in Python and R and can be applied to

Guo-Cheng Yuan (ed.), *Computational Methods for Single-Cell Data Analysis*, Methods in Molecular Biology, vol. 1935, https://doi.org/10.1007/978-1-4939-9057-3_5, © Springer Science+Business Media, LLC, part of Springer Nature 2019

datasets originating from different platforms, such as multiplex qPCR data, traditional single-cell RNA-seq, or UMI-based single-cell RNA-seq, e.g., inDrops, Drop-seq, and 10× genomics.

2 Materials

2.1 Operating System

Operating system is preferred to be Linux or MAC. In windows, Cygwin is highly recommended to build a Linux-like environment. It can be downloaded from https://cygwin.com/install.html. For the installation of GiniClust and its dependencies, Internet access is necessary.

2.2 Python Packages

The graphical user interface of GiniClust relies on wxPython, a Python wrapper for the cross-platform wxWidgets API. Please ensure that you have Python 2.7 in your environment. In addition, GiniClust relies on the following libraries:

1. Gooey.
2. Setuptools.

Those packages should be automatically installed or upgraded via a pip installation. For instance, to install Gooey, proceed as follows:

3. Start a terminal session;
4. `run $ pip install Gooey --upgrade.`

If in doubt, please check that those libraries got installed properly by trying to import them or some of their modules in your Python interpreter: `>>> import gooey, pkg_resources.`

2.3 R Packages

As for the R code at the core of much of GiniClust's computations, for MAC and WINDOWS only the official R (3.2.1 or higher version) installation file is supported and tested. Using other installation methods, such as brew, may lead to running error. Besides, some users might experience issues installing another of GiniClust's dependencies: the MAST R package (*see* **Note 1**).

2.4 Input Files

The input file is a gene expression matrix in comma-separated value (csv) format. Specifically, for qPCR data, each row is log2 gene expression level; for RNAseq data, each row is UMI-Count/Cell or Raw-Read-Count/Cell (*see* **Note 2**). The first row contains cell IDs. The first column contains unique gene names. For example, user can take a look at one of our test datasets (stored in the sample_data folder within GiniClust's repository):

```
>ExprM.RawCounts<-read.csv("Data_GBM.csv", sep=",", head=T)
>ExprM.RawCounts[1:4,1:4]
#            MGH26 MGH26.1 MGH26.2 MGH26.3
#1/2-SBSRNA4 0     47      0       0
#A1BG        41    80      3       0
#A1BG-AS1    0     0       0       0
#A1CF        0     0       0       0
```

3 Methods

3.1 Run GiniClust Through Python-Based Graphical User Interface

To run GiniClust, please download the GiniClust GitHub repository from https://github.com/lanjiangboston/GiniClust/archive/master.zip, unzip it and move to the extracted directory so that it becomes your current working directory. Then, in a Linux environment, start a terminal session and enter:

```
$ python GiniClust.py.
```

From an OS X or Windows environment, launch a terminal session and enter:

```
$ pythonw GiniClust.py
```

A graphical user interface springs up and directs you into choosing a file to process from your arborescence of directories, specify the type of data by choose "qpcr" or "rna" under the "Actions" column, along with the name of the folder where you would like to store GiniClust's output (*see* the section below for more information about those files). A screenshot is provided (Fig. 1).

3.2 Run GiniClust Through R Script Main Function

Alternatively, GiniClust can be run directly as an R script at the command-line interface. User can run GiniClust in terminal session using Rscript like following:

```
$ Rscript Giniclust_Main.R [options]
```

You can specify the following options:

- -f CHARACTER or --file = CHARACTER, input dataset file name
- -t CHARACTER or --type = CHARACTER, input dataset type: choose from 'qPCR' or 'RNA-seq'
- -o CHARACTER or --out = CHARACTER, output folder name [default = results]

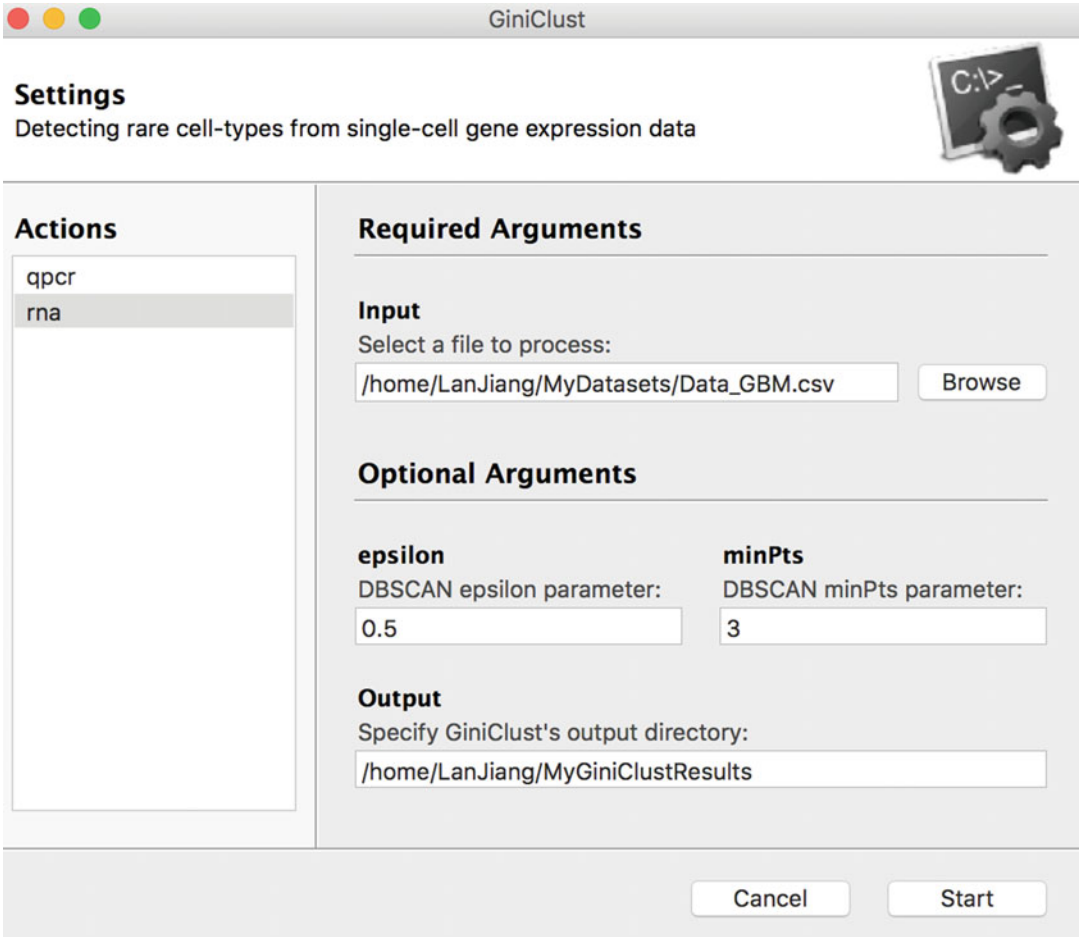

Fig. 1 Graphical user interface of GiniClust

- -e DOUBLE or --epsilon = DOUBLE, DBSCAN epsilon parameter qPCR:[default = 0.25],RNA-seq:[default = 0.5]
- -m INTEGER or --minPts = INTEGER, DBSCAN minPts parameter qPCR:[default = 5],RNA-seq:[default = 3]
- -h or --help, Show help message and exit.

For example, the following command is used to analyze the 'Data_GBM.csv' dataset.

```
$ Rscript GiniClust_Main.R -f Data_GBM.csv -t RNA-seq -o
GBM_results
```

The following command is used to analyze the 'Data_qPCR.csv' dataset.

```
$ Rscript GiniClust_Main.R -f Data_qPCR.csv -t qPCR -o
qPCR_results
```

Default parameters for GiniClust are shown as below:

```
minCellNum           = 3
minGeneNum           = 2000
expressed_cutoff     = 1
log2.expr.cutoffl    = 0
log2.expr.cutoffh    = 20
Gini.pvalue_cutoff   = 0.0001
Norm.Gini.cutoff     = 1
span                 = 0.9
outlier_remove       = 0.75
Gamma                = 0.9
diff.cutoff          = 1
lr.p_value_cutoff    = 1e-5
CountsForNormalized  = 100000
rare_p               = 0.05
perplexity           = 30
```

Usually default parameters work well for identifying rare cell types for datasets from different platforms. However, the user can always explore with different parameters. Users can use a text editor to open "yourworkdir/Rfunction/GiniClust_parameters.R." After changing parameters, users can rerun the GiniClust through Giniclust_Main.R script.

3.3 Run GiniClust Through R Script Step by Step

While GiniClust_Main.R is more friendly to use, running Giniclust step by step gives users more control and thus is more suitable for customized analysis. GiniClust stores intermediate files for each step. So it is convenient for users to check the results and to change parameters if necessary and rerun a single step before moving to the next one. A step-by-step instruction of running GiniClust is described in the following:

3.3.1 Loading Packages

This step first loads all the additional required packages. If it is the first time running GiniClust, this step automatically installs all R packages, so it is necessary to have Internet access.

```
>source("Rfunction/GiniClust_packages.R")
```

Additionally, users can load all the R functions defined by GiniClust:

```
> source("Rfunction/GiniClust_Preprocess.R")
> source("Rfunction/GiniClust_Filtering.R")
> source("Rfunction/GiniClust_Fitting.R")
> source("Rfunction/GiniClust_Clustering.R")
> source("Rfunction/GiniClust_tSNE.R")
> source("Rfunction/DE_MAST.R")
> source("Rfunction/DE_t_test.R")
```

3.3.2 Preprocessing

This step is used for preprocessing and loading the input data. The input data for RNA-seq should be raw read counts or UMI counts. For qPCR data, since the input data are in the log2 scale, GiniClust transforms back to normal scale to consistently process the data for the later steps. The variables of input include data.type, out.folder, and exprimentID, while the file of input is "exprimentID_rawCounts.csv." And the variable of output is "ExprM.RawCounts," and is saved as a file named "exprimentID_rawCounts.csv."

```
> ExprM.RawCounts = GiniClust_Preprocess(data.file, data.type, out.folder,
exprimentID)
```

3.3.3 Filtering the Preprocessed Data

GiniClust provides some basic filtering functions. The variable of input is "ExprM.RawCounts." The variable of output is "ExprM.RawCounts.filter." The intermediate stored file is "exprimentID_gene.expression.matrix.RawCounts.filtered.csv." The parameters involved and default values are:

```
> minCellNum          = 3
> minGeneNum          = 2000
> expressed_cutoff    = 1
```

After modifying the parameters (*see* **Note 3**), run the code below:

```
> ExprM.Results.filter=GiniClust_Filtering(ExprM.RawCounts, out.folder,
exprimentID)
```

3.3.4 Selecting a Subset of the Genes for Clustering

This step normalizes the Gini index by LOESS curve fitting in Gini vs. Max space. The variable of input is "ExprM.RawCounts.filter." The variable of output is "Genelist.top_pvalue" or "Genelist.HighNormGini." And the intermediate stored files are "Gini_related_table_RNA-seq.csv" or "Gini_related_table_qPCR.csv." The parameters involved and default values are:

```
> log2.expr.cutoffl      = 0
> log2.expr.cutoffh      = 20
> Gini.pvalue_cutoff     = 0.0001
> Norm.Gini.cutoff       = 1
> span                   = 0.9
> outlier_remove         = 0.75
```

After modifying the parameters (*see* **Note 4**), run the code below:

```
> GeneList.final = GiniClust_Fitting(data.type, ExprM.RawCounts.filter, out.folder,
exprimentID)
```

3.3.5 Clustering

This step builds a cell-cell discard distance based on selected genes from **step 4**. Then DBSCAN is used to detect clusters. The variable of input is "ExprM.RawCounts.filter." The variable of output is "cell.cell.distance,", "c_membership," "clustering_membership_r," and "rare.cells.list.all." And the intermediate files are "exprimentID _clusterID.csv," "exprimentID _rare_cells_list.txt." The parameters involved and default values are:

```
> eps       = 0.5
> MinPts    = 3
> rare_p    = 0.05
```

After modifying the parameters (*see* **Note 5**), run the code below:

```
> Cluster.Results =GiniClust_Clustering(data.type, \
                 ExprM.RawCounts.filter,\
                 GeneList.final,eps,\
                 MinPts, \
                 out.folder,\
                 exprimentID)
> cell.cell.distance       = Cluster.Results$cell_cell_dist
> c_membership             = Cluster.Results$c_membership
> clustering_membership_r = Cluster.Results$clustering_membership_r
> rare.cells.list.all      = Cluster.Results$rare.cell
```

Users can check the clustering results, for example,

```
> table(c_membership)
> print(rare.cells.list.all)
```

3.3.6 tSNE Visualization

A nonlinear dimension reduction technique called tSNE is used to visualize clustering results. The variable of input is "c_membership," "cell.cell.distance." The variable of output is "Rtnse_coord2." And the intermediate stored files are "exprimentID_Rtnse_coord2.csv." A figure is also generated and stored in the work folder (Fig. 2). The parameters involved and default values are:

```
> perplexity           = 30
```

After modifying the parameters (*see* **Note 6**), run the code below:

```
> GiniClust_tSNE(data.type, c_membership , cell.cell.distance, \
                 perplexity,  out.folder, exprimentID)
```

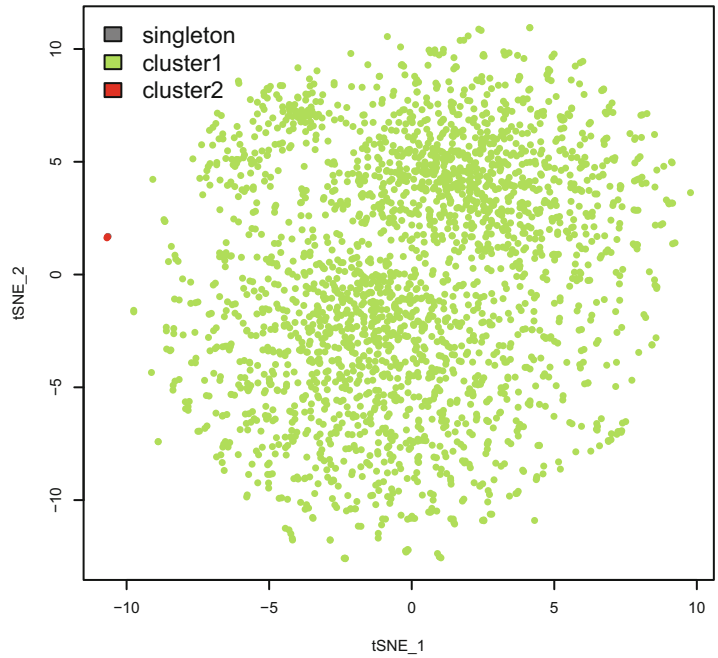

Fig. 2 An example of tSNE visualization of clustering results

3.3.7 Differentially Expressed Genes

Comparing the single-cell expression data of each putative rare cell type cluster to the largest cluster will identify differentially expressed genes. The variable of input is "ExprM.RawCounts.filter," "rare.cells.list.all," "c_membership." The variable of output is "differential.r." And the intermediate stored files are "RareCluster.overlap_genes.txt" and "RareCluster_lrTest.csv" or "RareCluster.diff.gene.t-test.results.csv." The parameters involved and default values are:

```
diff.cutoff          = 1
lr.p_value_cutoff    = 1e-5
```

After modifying the parameters (*see* **Note 7**), for RNA-seq data, run the code below:

```
> DE_MAST(ExprM.RawCounts.filter, rare.cells.list.all, out.folder, exprimentID)
```

For qPCR data, run the code below:

```
>DE_t_test(ExprM.RawCounts.filter,rare.cells.list.all,c_membership,out.folder,exprimentID)
```

3.4 Full List of Output Files and Description

The output directory specified by the user at the graphical user interface contains the following files and directories.

Main results:

exprimentID_rare_cells_list.txt: the clusters of rare cells detected by Giniclust.

RareCluster_lrTest.csv or RareCluster.diff.gene.t-test.results.csv: Differentially expressed genes results for the rare cells type cluster.

Other supporting results:

exprimentID_rawCounts.csv: the raw counts.

exprimentID_normCounts.csv: the normalized counts.

exprimentID_gene.expression.matrix.RawCounts.filtered.csv: the raw counts after filtering.

exprimentID_gene.expression.matrix.normCounts.filtered.csv: the normalized counts after filtering.

Gini_related_table_RNA-seq.csv: the table related with Gini index for RNA-seq data.

Gini_related_table_qPCR.csv: the table related with Gini index for qPCR data.

exprimentID_clusterID.csv: clustering result, the first column represents cell IDs and the second column is the corresponding cluster result for each cell.

exprimentID_Rtnse_coord2.csv: coordinates of cells in tSNE plot.

exprimentID_bi-directional.GiniIndexTable.csv: For qPCR data the table of bidirectional Gini index.

RareCluster.overlap_genes.txt: overlap genes between the selected high Gini genes and DE genes in rare cluster.

Sub-folder 'figures':

exprimentID_histogram of Normalized.Gini.Socre.pdf: histogram of estimated *p*-values based on a normal distribution approximation for genes.

exprimentID_smoothScatter_pvalue_gene.pdf: the smoothScatter plot in which the red points are the selected high Gini genes according to specified cutoff.

exprimentID_tsne_plot.pdf: tSNE plot for cells.

exprimentID_RareCluster_diff_gene_overlap.pdf: Venn diagram for differentially expressed genes and high gini genes.

exprimentID_RareCluster_overlapgene_rawCounts_bar_plot.genename.pdf: barplot of rare cluster and major cluster for the overlap genes.

3.5 Further Reading

It is worth noting that while the GiniClust is powerful for detecting rare cell types cluster, it is not sensitive for distinguishing major cell types. This limitation can be partially resolved by updated version of GiniClust called GiniClust2. It uses a novel cluster-aware weighted

consensus clustering algorithm to combine GiniClust and Fano-based k-means clustering results, by maximizing the strengths of these individual clustering methods in detecting rare and common clusters, respectively [4].

4 Notes

1. If users have a problem with installing MAST packages, please visit the website (https://github.com/RGLab/MAST) for detailed instructions. We recommend that users upgrade MAST package to the newest version. If you are using an old version, you may need to replace the file DE_MAST.R by https://github.com/lanjiangboston/GiniClust/blob/master/Archive/DE_MAST.R.
2. log2 transformed RNA-seq data for Giniclust may not work. We suggest that users use featureCounts from http://subread.sourceforge.net/ [5] or htseq-count from http://www-huber.embl.de/users/anders/HTSeq/doc/counting.html [6] to get raw reads counts.
3. minCellNum means the minimum number of cells for rare cell clusters. It is highly recommended that this value is set to be equal to or larger than 3. However, the larger value of this parameter means less sensitivity. minGeneNum is used for filtering cell that may not express enough genes. The default value of 2000 is consistent with recent report about scRNA-seq using smFISH as guide [7].
4. "log2.expr.cutoffl' " and "log2.expr.cutoffh" define the range of gene expression. "Gini.pvalue_cutoff" controls how many genes finally are chosen. "Norm.Gini.cutoff" controls choosing high gini genes based on p-value or not (default = 1). "Span" and "outlier_remove" are parameters used in LOESS fitting.
5. "eps" and "MinPts" are parameter of "epsilon" and "minimum points" for DBSCAN, respectively. The distance of any point to its nearest core point of the same cluster is less than "epsilon." larger values for "MinPts" are usually better for data sets with noise and form more significant clusters, and however, lose the sensitivity for rare cell clusters. "rare_p" is a parameter to define what you call rare cell cluster. For example, rare_p = 0.05 means only cluster that contributes to less than 5% of the whole population is called rare cell cluster.
6. "Perplexity" is a parameter for tSNE. The results of t-SNE are fairly robust for different perplexity. The most appropriate value depends on the density and size of the data. Typical values for the perplexity range between 5 and 50.

7. "diff.cutoff" and "lr.p_value_cutoff" are used to filter differential expressed genes based on log2 fold change and *p* value during MAST analysis, respectively. Increasing the value of "diff.cutoff" or decreasing the value of "lr.p_value_cutoff" will decrease the number of differential expressed genes.

References

1. Habib N, Li Y, Heidenreich M, Swiech L, Avraham-Davidi I, Trombetta JJ, Hession C, Zhang F, Regev A (2016) Div-Seq: single-nucleus RNA-Seq reveals dynamics of rare adult newborn neurons. Science 353 (6302):925–928. https://doi.org/10.1126/science.aad7038
2. Sharma SV, Lee DY, Li B, Quinlan MP, Takahashi F, Maheswaran S, McDermott U, Azizian N, Zou L, Fischbach MA, Wong KK, Brandstetter K, Wittner B, Ramaswamy S, Classon M, Settleman J (2010) A chromatin-mediated reversible drug-tolerant state in cancer cell subpopulations. Cell 141 (1):69–80. https://doi.org/10.1016/j.cell.2010.02.027
3. Jiang L, Chen H, Pinello L, Yuan GC (2016) GiniClust: detecting rare cell types from single-cell gene expression data with Gini index. Genome Biol 17(1):144. https://doi.org/10.1186/s13059-016-1010-4
4. Tsoucas D, Yuan GC (2018) GiniClust2: a cluster-aware, weighted ensemble clustering method for cell-type detection. Genome Biol 19(1):58. https://doi.org/10.1186/s13059-018-1431-3
5. Liao Y, Smyth GK, Shi W (2014) Featurecounts: an efficient general purpose program for assigning sequence reads to genomic features. Bioinformatics 30(7):923–930. https://doi.org/10.1093/bioinformatics/btt656
6. Anders S, Pyl PT, Huber W (2015) HTSeq--a Python framework to work with high-throughput sequencing data. Bioinformatics 31 (2):166–169. https://doi.org/10.1093/bioinformatics/btu638
7. Torre E, Dueck H, Shaffer S, Gospocic J, Gupte R, Bonasio R, Kim J, Murray J, Raj A (2018) Rare cell detection by single-cell RNA sequencing as guided by single-molecule RNA FISH. Cell Syst 6(2):171–179.e175. https://doi.org/10.1016/j.cels.2018.01.014

ExLibris RapidILL

Rapid #: -20454510

CROSS REF ID: **815163**

LENDER: **ORE :: Main Library**

BORROWER: **VWM :: Swem Library**

TYPE: Book Chapter

BOOK TITLE: Computational Methods for Single-Cell Data Analysis

USER BOOK TITLE: Computational Methods for Single-Cell Data Analysis

CHAPTER TITLE: Rare Cell Type Detection

BOOK AUTHOR: Jiang

EDITION:

VOLUME:

PUBLISHER: Springer

YEAR: 2019

PAGES: 79-89

ISBN: 9781493990573

LCCN:

OCLC #:

Processed by RapidX: 3/16/2023 10:52:20 AM

This material may be protected by copyright law (Title 17 U.S. Code)